REPLICATION

Learning Outcome

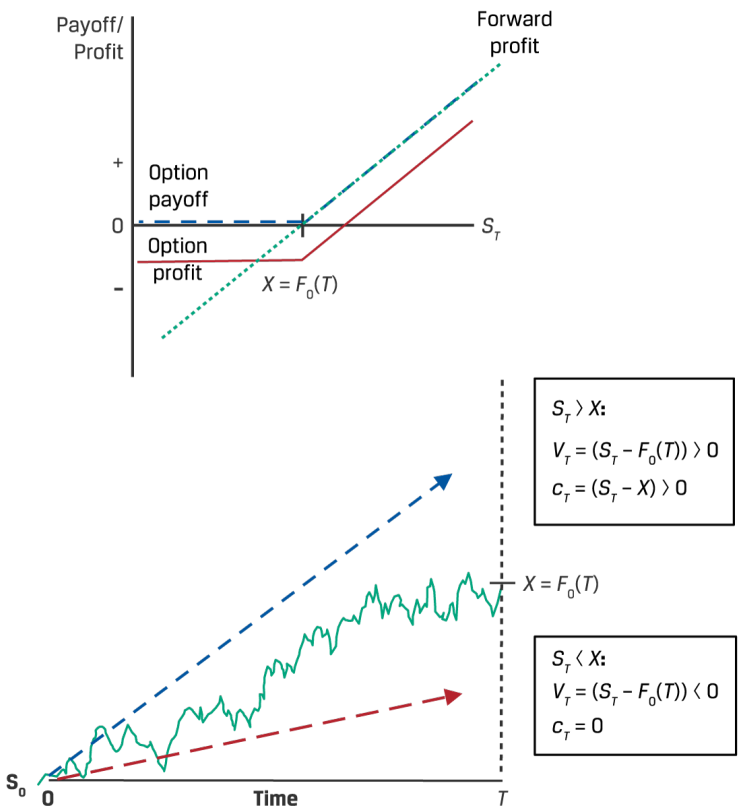
- contrast the use of arbitrage and replication concepts in pricing forward commitments and contingent claims

In an earlier lesson, we used replication to create forward commitment cash flows, utilizing a combination of long or short positions in an underlying asset and borrowing or lending cash. The ability of market participants to use replication strategies ensures that the law of one price holds and no riskless arbitrage profit opportunities exist.

Recall from an earlier lesson that a call option is *similar to* a long forward position in that it increases in value as the underlying price rises but *differs* in that the call option settles only if there is a gain upon exercise. This apparent symmetry for positive outcomes is shown in Exhibit 4, where the exercise price, X, is equal to the forward price, $F_0(T)$.

Exhibit 4:

Call Option versus Long Forward Position



▶ Expand / collapse Extended Description

In order to replicate the call option at contract inception ($t = 0$), we also must borrow at the risk-free rate, r, and use the proceeds to purchase the underlying at a price of S_0. At option expiration ($t = T$), unlike in the case of the forward commitment, there are two possible replication outcomes:

- **Exercise** ($S_T > X$): Sell the underlying for S_T and use the proceeds to repay X.
- **No exercise** ($S_T < X$): No settlement is required.

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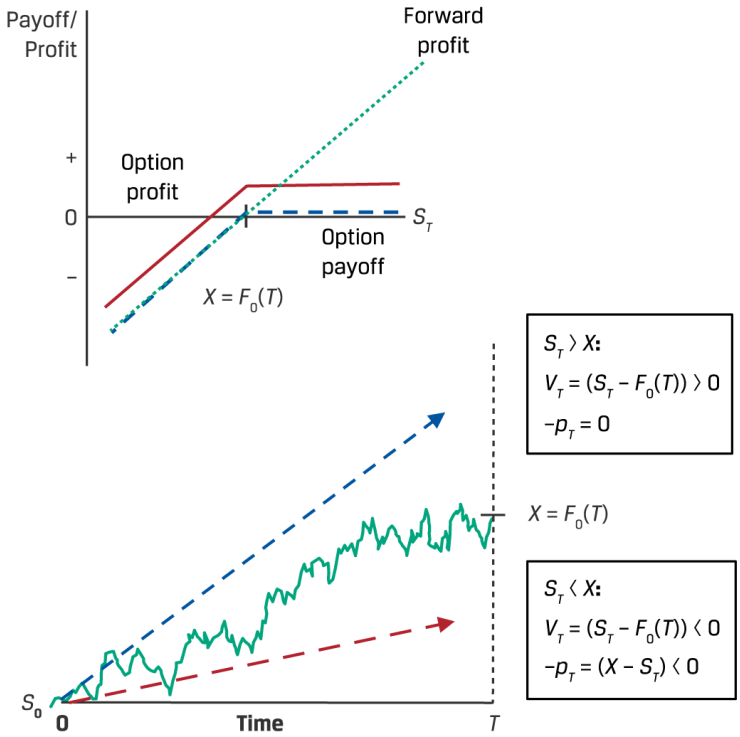
on the *likelihood* of exercise at time T, which is closely associated with the moneyness of an option. The non-linear payoff profile of an option requires that the replicating transaction be adjusted as this likelihood changes, while the replicating trades for a forward commitment remain constant. Option replication will be addressed in greater detail in later lessons.

We now turn our attention to put option replication. It was shown earlier that a *short* put position with an exercise price equal to the forward price ($X = F_0(T)$) mirrors the outcomes for a *long* forward position when the underlying price at maturity is *below* the forward price ($S_T < X = F_0(T)$). The sold put *decreases* in value as the underlying price falls, but it settles only if there is a gain upon exercise. This relationship is shown in Exhibit 5, where the exercise price (X) is equal to the forward price, $F_0(T)$.

Exhibit 5:

Short Put Option versus Long Forward Position



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Accordingly, as a short put option would require a long position in the underlying for replication, so in order to replicate a long position in the put option at contract inception ($t = 0$), we must sell the underlying short at a price of S_0 and lend the proceeds at the risk-free rate, r. At option expiration ($t = T$), unlike in the case of the forward commitment, there are two possible replication outcomes:

- **Exercise** ($S_T < X$): Purchase the underlying for S_T from the proceeds of the risk-free loan.
- **No exercise** ($S_T > X$): No settlement is required.

If exercise were certain, we would lend $X(1 + r)^{-T}$ at inception, as in the case of the forward. However, as in the case of the call option, the asymmetric payoff profile requires adjustment over time based on the *likelihood* of exercise.

In the next lesson, we will turn our attention to the factors that drive the likelihood of option exercise prior to maturity.

1. Determine the correct answers to complete the following sentence: The lower bound of a call price is the underlying's price _____ the present value of its _____ price or zero, whichever is greater.

Hide Solution

Solution:

The lower bound of a call price is the underlying's price *minus* the present value of its *exercise* price or zero, whichever is greater.

2. A six-month European call option on an underlying stock with no additional cash flows has an exercise price (X) of GBP 50, an initial underlying price (S_0) of GBP 49.75, and a risk-free rate of 2%. Calculate the lower bound of the call price in three months' time if S_t = GBP 65.

Hide Solution

Solution:

As the call option buyer will exercise only if $S_T > X$ at maturity, the lower bound is equal to S_t − PV (X) or zero in three months' time, whichever is greater:

$$c_{t,Lower\ bound} = Max\left(0,\ S_t - X(1+r)^{-(T-t)}\right)$$

Given S_t = GBP 65, X = GBP 50, r = 2%, and (T − t) = 0.25, we can solve for the call option's lower bound as follows:

$$c_{t,Lower\ bound} = Max\left(0,\ \text{GBP65} - \text{GBP50}(1.02)^{-(0.25)}\right)$$

= GBP 15.25

3. Match the following statements about replication strategies with their associated derivative instrument(s):

1. At time t = 0, borrow at the risk-free rate and purchase the underlying at S_0.	A. Both a call option and a put option replication strategy
2. The strategy requires adjustment over time as the likelihood of exercise changes.	B. A call option replication strategy
3. At time t = T, receive the loan repayment and purchase the underlying at S_T.	C. A put option replication strategy if exercised

Hide Solution

Solution:

1. B is correct. At time t = 0, a call option replication strategy involves borrowing at the risk-free rate and purchasing the underlying.
2. A is correct. As both call and put options have a non-linear payoff profile, the replication strategy requires adjustment over time as the likelihood of exercise changes.
3. C is correct. A put option replication strategy if exercised involves receiving the loan repayment and purchasing the underlying at S_T at time t = T.

4. Explain the key difference between the changes in the replication of a contingent claim versus a forward commitment over the life of the contract.

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exercise changes over time, transactions replicating an option contract must be adjusted over time. A forward commitment has a *symmetric* payoff profile that will settle with certainty at a future date, and therefore the replicating transactions do not require adjustment over the contract life.

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